

11 (a) A medical centre uses objects of the class Appointment to record treatments given and medication prescribed during each doctor's appointment. Some of the attributes required in the class are listed in the table.

Attribute	Data type	Description
DateSeen	DATE	date of treatment
Treatments	STRING	treatments given
Medications	STRING	medications prescribed

Patients are identified by a unique 8-digit number, beginning with the patient's year of birth, for example, 20108989.
Doctors are identified by their name, for example, A N Other.

Complete the class diagram for Appointment, to include:

- attribute and data type for the identification of the patient
- attribute and data type for the identification of the doctor
- methods to assign date seen, treatments given and medications prescribed
- method to return the date seen and the attributes for the patient and the doctor.

Appointment	
DateSeen	: DATE
.....	:
.....	:
Treatments	: STRING
Medications	: STRING
.....	
SetPatientID(PatientNumber : INTEGER)	
SetDoctor(DoctorID : STRING)	
.....	
.....	
.....	
.....	
.....	
GetTreatments()	
GetMedications()	

[5]

(b) (i) Identify the object-oriented programming (OOP) feature whose function includes restricting external access to the data.

..... [1]

(ii) Describe what is meant by the OOP feature inheritance.

.....
.....
.....
..... [2]

12 The pseudocode algorithm checks whether a location in a stock file StockList.dat is empty or not. The location is given by the user. If the location is empty, a suitable message is displayed, otherwise the item stored at that location is displayed.

Complete this file-handling pseudocode algorithm.

```
DECLARE Location : INTEGER
DECLARE Item : STRING
DECLARE Continue : BOOLEAN
DECLARE Answer : CHAR
Continue ← TRUE

OPENFILE .....
WHILE Continue
    OUTPUT "Enter a location between 1 and 500: "
    INPUT Location

    .....

    GETRECORD .....
    IF Item = "" THEN
        OUTPUT "This record is missing."
    ELSE
        OUTPUT "The item in stock is ", .....
    ENDIF
    OUTPUT "Another location (Y or N)?"
    INPUT Answer
    IF Answer <> 'Y' THEN
        Continue ← FALSE
    ENDIF
ENDWHILE

.....
OUTPUT "End of program"
```

[5]